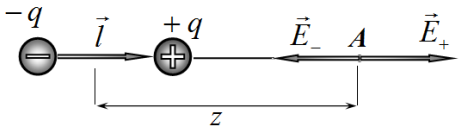
제2 부: 전기

2.1.1



$$E = E_+ - E_- = \frac{q}{4\pi\varepsilon_0\left(z-\frac{l}{2}\right)^2} - \frac{q}{4\pi\varepsilon_0\left(z+\frac{l}{2}\right)^2} = \frac{q\cdot 2zl}{4\pi\varepsilon_0\left(z^2-\left(\frac{l}{2}\right)^2\right)^2} = \frac{zp_e}{2\pi\varepsilon_0\left(z^2-\left(\frac{l}{2}\right)^2\right)^2} \quad (19)$$

$z >> l$

$$E = \frac{zp_e}{2\pi\varepsilon_0\left(z^2-\left(\frac{l}{2}\right)^2\right)^2} = \frac{1}{2\pi\varepsilon_0}\frac{p_e}{z^3} \quad (20)$$

$$a = \frac{1}{2\pi\varepsilon_0}, \qquad \alpha = 3 \quad (21)$$

2.2.1 .

. $\vec{E}_0$

q $\pm\sigma$.

$\vec{E}'$.

,

$\vec{E}_0$. ,

.

$$E_0 = \frac{q}{4\pi\varepsilon_0 z^2},$$

$$E' = \frac{\sigma}{\varepsilon_0}.$$

$$\sigma' = \frac{q}{4\pi z^2}$$

$$q' = \sigma' S = \frac{qS}{4\pi z^2}, \qquad S = \pi R^2$$

.

$$p'_e = q'h = \frac{qSh}{4\pi z^2} = \frac{q}{4\pi z^2}V. \quad (22)$$

$$F = qE = q\frac{1}{2\pi\varepsilon_0}\frac{p'_e}{z^3} = \frac{q^2}{8\pi^2\varepsilon_0 z^5}V \quad (23)$$

, z $(z-x)$. $x << z$

$$F = \frac{q^2V}{8\pi^2\varepsilon_0(z-x)^5} = \frac{q^2V}{8\pi^2\varepsilon_0 z^5}\left(1-\frac{x}{z}\right)^{-5} \approx \frac{q^2V}{8\pi^2\varepsilon_0 z^5} + 5\frac{q^2V}{8\pi^2\varepsilon_0 z^6}x \quad (24)$$